

HR Analytics Dashboard

A Business Analysis Case Study

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Project Overview

This project involved designing an HR analytics dashboard to help leadership understand and act on employee attrition. The goal was to consolidate workforce data — attrition, department, region, job role, overtime, job satisfaction, performance, salary, and training — into a single-page dashboard answering the ten most pressing questions HR and business leaders ask about retention.

Business Objective

The organization needed to understand why attrition was high and where it was concentrated, so HR could prioritize retention efforts instead of applying generic, organization-wide interventions. The dashboard was designed to answer questions such as: What is the overall attrition rate? Which departments, regions, and roles are most affected? Does overtime drive attrition? How do job satisfaction, performance, and salary relate to retention? And what actions would most effectively reduce attrition?

Key Workforce Metrics

ATTRITION RATE 44.0%	AVG SALARY \$53.9K	AVG TRAINING HOURS 38.0	AVG JOB SATISFACTION 3.0 / 5
AVG PERFORMANCE RATE 3.1 / 5	OT EMPLOYEES 256	TOTAL EMPLOYEES 500	

Approach & Methodology

- **Data consolidation:** Combined attrition, department, region, job role, overtime, satisfaction, performance, salary, and training data into one analytical view.
- **Segmentation analysis:** Calculated overall and segmented attrition rates by department, region, job role, overtime status, and satisfaction level to isolate the biggest risk factors.
- **Cross-factor analysis:** Cross-tabulated job role against satisfaction level to identify the highest-risk employee groups, not just single-variable drivers.
- **Dashboard design:** Structured the dashboard around business questions, insights, and recommendations so it functions as a decision-support tool for HR leadership.
- **Insight-to-action mapping:** Converted each insight into a specific, actionable recommendation aimed at the relevant department, region, or role.

Key Insights

- **High overall attrition:** The organization's overall attrition rate is 44.0% (220 employees) — well above a healthy benchmark of 10–15%, signalling a significant retention challenge.
- **Departmental risk:** IT (51) and HR (45) have the highest attrition, while Sales and Operations (27 each) have the lowest.
- **Regional risk:** Gauteng (48) and Eastern Cape (47) show the highest attrition, while Limpopo (40) has the lowest.

- **Overtime impact:** Employees who work overtime show noticeably higher attrition than those who do not, pointing to possible burnout risk.
- **Role-level risk:** Assistants (51) and Analysts (48) have the highest attrition by job role, followed by Managers (43).
- **Satisfaction pattern:** Attrition is highest among employees with very low (level 1) or high (level 4) job satisfaction, and lowest at satisfaction level 5.
- **Combined risk group:** Assistants with low satisfaction (levels 1–2) show the highest attrition of any group (15 at level 1, 11 at level 2).
- **Training gap:** Average training hours are 38.0, which may be insufficient to meaningfully drive engagement and retention.
- **Salary is not the sole driver:** Average salary is \$53.9K, but salary alone does not explain attrition, since high attrition persists across multiple roles and regions.

Recommendations

- **Prioritize high-risk departments:** Focus immediate retention efforts — engagement initiatives, leadership support, and career development — on IT and HR.
- **Regional retention strategy:** Investigate root causes in Gauteng and Eastern Cape and implement region-specific action plans, including manager training and engagement programs.
- **Manage overtime effectively:** Review workloads and staffing levels for overtime employees, and provide flexibility where possible to reduce burnout risk.
- **Support assistants and analysts:** Provide clear career paths, skill development, mentorship, and recognition for Assistants and Analysts, the highest-attrition roles.
- **Improve job satisfaction:** Conduct surveys to understand pain points — work-life balance, recognition, management — driving attrition at low and high satisfaction extremes.
- **Target low-satisfaction groups:** Design tailored interventions and feedback sessions for employees at satisfaction levels 1–2, especially Assistants.
- **Increase employee development:** Increase training hours above the current 38.0 average and align content with career growth and role requirements.
- **Strengthen engagement and recognition:** Introduce regular recognition, feedback loops, and engagement activities to strengthen satisfaction and commitment.
- **Monitor and track progress:** Set monthly and quarterly attrition targets and track progress by department, region, role, and satisfaction level.

Key Takeaway

The organization's 44.0% attrition rate is well above a healthy benchmark, concentrated in IT and HR, in the Gauteng and Eastern Cape regions, and among Assistants and Analysts. Overtime and low job satisfaction are strongly linked to higher attrition. Focused retention strategies, employee development, workload management, and engagement initiatives are critical to stabilizing the workforce.

Tools & Skills Demonstrated

- HR and workforce analytics dashboard design
- Attrition analysis and employee segmentation
- Cross-factor analysis (role × satisfaction, overtime × attrition)
- Data visualisation and KPI reporting
- Translating data insights into actionable HR recommendations